FloodSight

Flood damage and cut-off settlements from raw Sentinel data, for any user-selected area and date

Multimodal AI Hackathon 2026 | Track B: Mapping Flood Damage from Space

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Technical report | Case study: Trishuli valley, Nepal, vs Copernicus EMS EMSR927

Educational prototype built only from allowed inputs; reference damage maps are used for checking only.

96%	98%	+0.17	0.55 / 0.84	9	28
of what we flag is also damaged on the official EMS map (Trishuli)	of the road length we close is damaged on the EMS map (Trishuli, EMS area 3)	better overlap score (IoU) than radar rules alone, on floods the AI never trained on	AI-alone overlap score (IoU) on an unseen 2026 flood, Crete (latest / first EMS map)	methods compared under a rule fixed before training	settlements found cut off from a hospital

What it is. FloodSight maps new flood water and debris for an area and date chosen by the user, starting from raw Sentinel-1 radar (Level-1 GRD) and adding Sentinel-2 optical images, the Copernicus DEM (terrain heights) and the pre-event OpenStreetMap (roads, buildings, hospitals). A Siamese U-Net trained only on the listed Kuro Siwo flood dataset estimates flood-water probability; physics-based radar rules (which also detect debris), terrain constraints and optical evidence are fused with it, and every detection is labelled **confirmed** (two independent sources inside the system agree) or **possible** (one source). FloodSight then lists the buildings, roads and bridges hit and the settlements that lost their road to a hospital, and produces a dashboard, a one-page situation report and GIS files.

Evidence. On the Trishuli case study the final map has precision 0.958 (96% of what it flags is also damaged on EMSR927, the official Copernicus Emergency Management Service map of this flood) and IoU 0.466 (overlap: shared area divided by combined area, 1 = perfect), and 98% of the road length FloodSight closes in EMS area 3 is damaged in EMS. The AI beats radar rules alone by **+0.17 IoU** on held-out Kuro Siwo events (0.653 vs 0.479, same tiles), was kept over eight alternatives by a rule fixed before training, and was tested blind on a second dataset (Sen1Floods11), a Himalayan flash flood (Buner) and a 2026 flood (Crete). Reference maps are used only to score results, never by the system.

Main limitation. Debris that changes the radar signal little, and damage narrower than the 20 m grid, are often missed in steep valleys (Himalayan recall 10-48%). FloodSight therefore separates confirmed, possible and radar-unknown areas instead of treating anything undetected as safe.

1. Problem and approach

After a Himalayan flash flood, responders need three answers fast: where the water and debris are, what infrastructure is damaged, and which villages are cut off. Monsoon clouds hide optical satellites, mountains distort radar, and debris is harder to see than open water. FloodSight answers the three questions from free Copernicus data for **any user-selected area and date** (subject to Sentinel coverage; accuracy varies with terrain, Section 6), with one command or one click. The same code and settings run in mountain valleys and on plains; nothing is set per area. Figure 1 shows the pipeline.

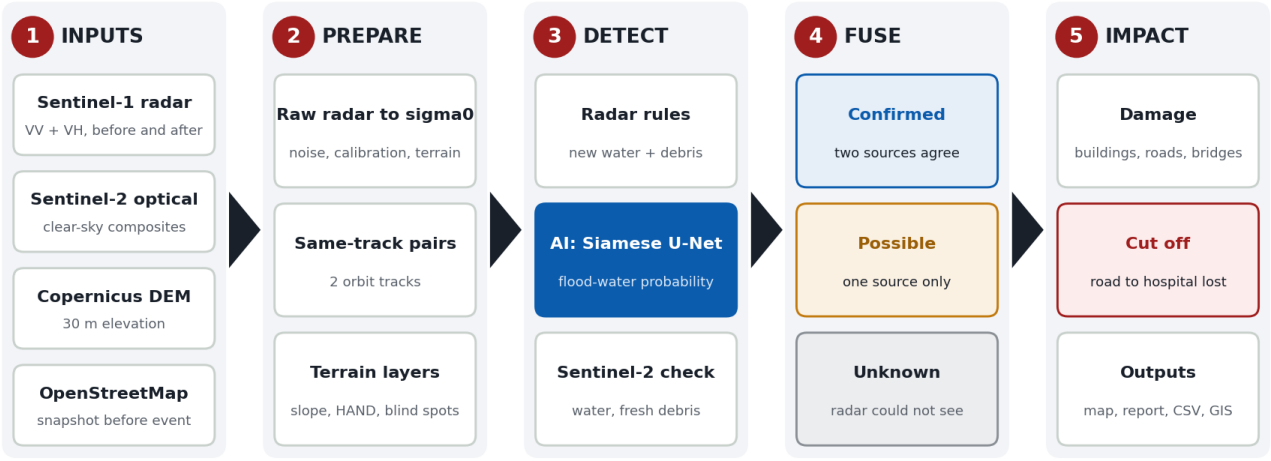


Figure 1. FloodSight in five stages. Reference maps (Copernicus EMS) are never an input; the code refuses to read them outside the validation module.

2. Data and raw-data processing

Input (allowed list)	Use	Rule
Sentinel-1 IW GRDH, VV+VH	Main detector of water and debris	Same-orbit-track pair; latest before, earliest after
Sentinel-2 L2A	Second sensor where the sky is clear	Median before (45 d), first clear after (20 d); SCL cloud/shadow/snow mask
Copernicus DEM GLO-30	Slope, drainage, HAND, radar blind spots, flow path	Static
OpenStreetMap (ohsome API)	Buildings, roads, bridges, settlements, hospitals	Snapshot strictly before the event
Kuro Siwo; Sen1Floods11	Train the AI; second test dataset (never trained on)	Splits by flood event

Image pairs. The CDSE catalogue is searched 30 days before to 14 days after the date. Only dual-polarisation (VV+VH) frames are kept. For each orbit track the latest image before the event and the earliest after it form a pair, so both were taken from the same viewing geometry and a change in brightness is a change on the ground. Up to two tracks (typically ascending and descending) are processed. For floods that build up over weeks the operator can give the last dry day (--before-date); all before inputs then come from before it.

From Level-1 to calibrated backscatter. GRD data are digital numbers in radar geometry with thermal noise. FloodSight's own chain reads only the window over the area, removes thermal noise with the product's range-azimuth noise tables (power = DN^2 minus noise, as in ESA SNAP), calibrates to sigma nought with the product's look-up table, geocodes with the geolocation grid and corrects terrain parallax with the DEM. By default the same steps run server-side on the Copernicus Data Space (Sentinel Hub, SIGMA0_ELLIPSOID, on our grid); openEO is a fallback. Both images of a pair always use the same route. Sigma0 was chosen because the Kuro Siwo training images match it within 0.5 dB on flat ground (gamma0-terrain differs by 1.3 dB), measured on input statistics only, and the same calibration is used in training and deployment. The processing chains still differ: on event 205 the model scores IoU 0.58 on Kuro Siwo's own images of the area and 0.44 on ours. All layers share one UTM grid at 20 m (areas up to 3,000 km^2).

3. Detection method

3.1 Terrain

From the DEM we compute slope, D8 flow direction after filling pits and depressions, flow accumulation (streams start at 2 km^2 upstream area) and **HAND**, the height of each pixel above the stream it drains to (own implementation that also handles flow leaving the area, so plains are not discarded). Layover, shadow and cast shadow are computed per track from the satellite heading and incidence angle; pixels a track cannot see are 'unknown', never 'safe', and a second track fills them where it can. A published mountain definition (UNEP-WCMC, Kapos et al. 2000) decides where the debris class is allowed, so harvested fields on plains are not called debris.

3.2 Radar rules

After a 5 x 5 Lee speckle filter, **new flood water** is a pixel within the water zone (HAND at most 20 m, slope at most 12°) that is below a scene-adaptive water threshold after the flood but not before, and darkened by more than 3 dB. The threshold is computed for every scene: tiles with a clearly bimodal histogram (Ashman's D > 2) are pooled and Otsu's method applied (split-based approach of Martinis et al.; Chini et al. 2017), clamped to -24 to -15 dB. **Debris and mud** needs two of three independent signs (VV change > 3 dB, VH change > 3 dB, texture increase > 1.6x) within HAND 45 m and slope 25°, in mountain terrain and within 600 m of the river corridor. Objects below 4,800 m^2 are removed. Debris detection is **rule-based, not learned**: the allowed training data label water only. These are physical defaults fixed on 30 September before any real data was seen.

3.3 AI model

A **Siamese U-Net** with a ResNet-34 encoder (25.2 M parameters, ImageNet start) receives the before and after images through the same shared encoder; at every scale the decoder sees the after-image features and the absolute feature difference, so it learns what changed. Input per date: VV and VH in dB and terrain slope. It was trained on Kuro Siwo split **by flood event** (train 1111004, 1111005, 1111009, 411, 562; validation 1111013; test 427 and 1111006; 205, 421 and 498 never used) with weighted cross-entropy plus Dice loss, AdamW and a one-cycle schedule on a laptop GPU. The epoch and the 0.8 probability threshold were chosen on the validation event only. It runs on every orbit track in overlapping tiles; per pixel the highest probability among tracks that see the pixel is kept.

3.4 Fusion and confidence

Where the AI agrees with a radar flood pixel, confidence rises by one level; where it clearly disagrees ($p < 0.2$) it falls by one; it adds water the rules missed only where water can pond, at low confidence; it never changes debris, because its labels are

water only. Debris from one image starts at low confidence (its VV, VH and texture signs share one acquisition and can all come from rain-soaked soil) and is **confirmed only by independent evidence**: the other orbit track within 40 m or Sentinel-2. Sentinel-2 (MNDWI for water, NDVI loss for fresh debris, only where clear before and after) confirms, contradicts (green vegetation after the event), fills radar blind spots and adds what only it sees at low confidence. Confirmed therefore means agreement of independent evidence inside the system; EMS maps are never used by it. Results lead with **confirmed** numbers; single-source detections are shown as '+ possible', and if nothing confirmed exceeds 0.05 km² the verdict is 'possible flooding, not confirmed'.

3.5 Key design decisions and the evidence behind them

Decision	Why	Evidence
Change detection with a before-image (not after-image segmentation)	Dark ground that was already dark (rivers, shadow, tarmac) is not new damage	Removing the before-image costs about 0.2 test IoU (0.65 to 0.47)
AI fused with physics rules, not used alone	The AI learned water on plains; rules cover debris and remove physically impossible water	AI +0.17 IoU over rules on test; rules carry debris where the AI finds 0.6% (Buner)
Confirmed only when two independent sources agree	One image's VV, VH and texture can all be caused by rain-soaked soil	Pre-registered rule; confident-debris precision Buner 12% to 60% and 50% to 94%
Two orbit tracks, not one or four	Second viewing side halves radar blind spots and allows corroboration	Unknown area in Trishuli 0.6%; four tracks gave no gain at twice the time
Debris class only in mountain terrain (published definition)	Debris flows need steep catchments; plains floods appear as water	Harvested fields on the Mura plains no longer called debris; no threshold invented
Scene-adaptive water threshold	Radar brightness of water differs per scene and incidence angle	Computed per scene (e.g. -15 dB in both tracks of one run), clamped to physical limits
20 m analysis grid	Cost and speed; 10 m did not help in narrow valleys	Meža IoU 0.014 to 0.020 at four times the processing units

4. Damage, cut-off settlements and access

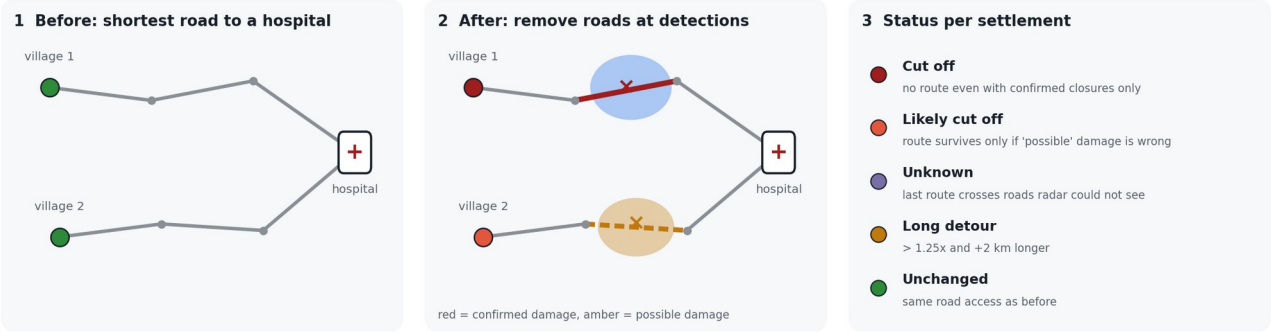


Figure 2. How a settlement is judged cut off: shortest road route to a hospital before and after removing roads at detected damage, separately for confirmed and possible damage.

- **Buildings** (pre-event outlines): likely hit at 30% of the footprint in confirmed flood/debris, possibly hit within 15 m or on low-confidence detections. **Roads** are sampled every 10 m and affected within 10 m of flood/debris (bridges 30 m); confirmed and possible lengths are kept separately.
- **Cut off:** a road graph is built from OpenStreetMap, 15 km beyond the area so escape routes are not cut by the map edge. For each settlement the shortest road distance and travel time (OSRM default speeds) to the nearest hospital are compared before and after under three closures: confirmed detections only, all detections, and roads the radar could not see. Status: cut off (no route even with confirmed closures), likely cut off, unknown, long detour (> 1.25x and +2 km), unchanged. A priority list ranks by certainty, size (mapped buildings) and extra time.
- **Air access** for cut-off settlements: mapped helipads with their state, and candidate open ground within 5 km (slope at most 7° per US Army FM 3-21.38, 60 m clear, away from flood, buildings and power lines), labelled as candidates for a pilot, not certified sites.
- **Bonus, flood path:** from any clicked point the path is traced downstream on the DEM (window by window, up to 120 km), listing settlements, hospitals and bridges with height above the river; it is drawn solid where it follows an OpenStreetMap river and dashed where it does not.

5. Evaluation protocol

Every setting is chosen on data that is not used to report the result it affects, and every choice is dated in a public tuning log. Tests are decided before they are run and reported as they come out; experiments state their success criterion in advance (two were rejected by their own criteria, Section 8). Against Copernicus EMS we rasterise the observed-event polygons

inside each EMS area of interest and report IoU, precision and recall (radar-blind pixels reported separately), using the latest map per area, plus confirmed versus possible precision, AI-alone and rules-alone scores, building matches and **road-closure precision**.

6. Results

How to read the scores. Precision: share of what FloodSight flags that the reference map also shows (false alarms lower it). **Recall**: share of the reference damage that FloodSight finds (misses lower it). **IoU**: overlap, shared area divided by combined area, from 0 to 1. An **EMS area** is one area that Copernicus EMS mapped, numbered by EMS. **Blind** = decided before running, never used to choose any setting; the case study was viewed during development (no setting chosen from it), so it is **not** blind. Rows score the final map, full pipeline from raw data.

Flood (reference)	Test type	IoU	Precision	Recall	Notes
Trishuli, Nepal, Aug 2026 (EMS EMSR927, all EMS areas)	Case study (not blind)	0.466	0.958	0.476	28 cut off, 5 likely; 98% of closed road length damaged in EMS (EMS area 3); 0.6% radar-unknown
Lasithi, Crete, Oct 2026 (EMS EMSR933, EMS area 3, latest map)	Blind	0.424	0.467	0.822	Confirmed detections P 0.72; first EMS map: IoU 0.476, confirmed P 0.85; 85% of EMS-damaged road found
Buner, Pakistan, Aug 2025 (EMS EMSR838 EMS areas 10 / 11)	Blind, Himalaya, run once	0.197 / 0.089	0.32 / 0.58	0.34 / 0.10	Confirmed P 0.60 / 0.94; debris-dominated flash flood
Sri Lanka, May 2017 (Kuro Siwo 205 labels)	Blind, held-out event	0.468	0.993	0.469	Never used in training or selection
Sindh, Pakistan, 2022 (Kuro Siwo 1111009)	Training event	0.848	-	-	Chain check, not a test of generalization

6.1 Case study: Trishuli valley vs EMSR927

The case-study box (85.08-85.40 E, 27.85-28.29 N) was chosen only to cover the EMS areas. Its scores were viewed during development, so it is not a blind test, but **no setting was ever chosen from them**; the blind tests are Crete, Buner, event 205 and Sen1Floods11. FloodSight flags damage with 96% precision and recovers 48% of the EMS mass-movement area; most misses are fresh debris on an already rough riverbed, which shows little radar change, and strips narrower than three 20 m pixels. The two orbit tracks leave only 0.6% of the valley unseen. 28 settlements are cut off and 5 likely cut off, and 98% of the road length FloodSight closes in EMS area 3 is damaged in EMS, so the cut-off analysis rests on reliable closures. EMS mapped this event as mass movement only, so the water-only AI is scored on the final map here and on its own elsewhere.

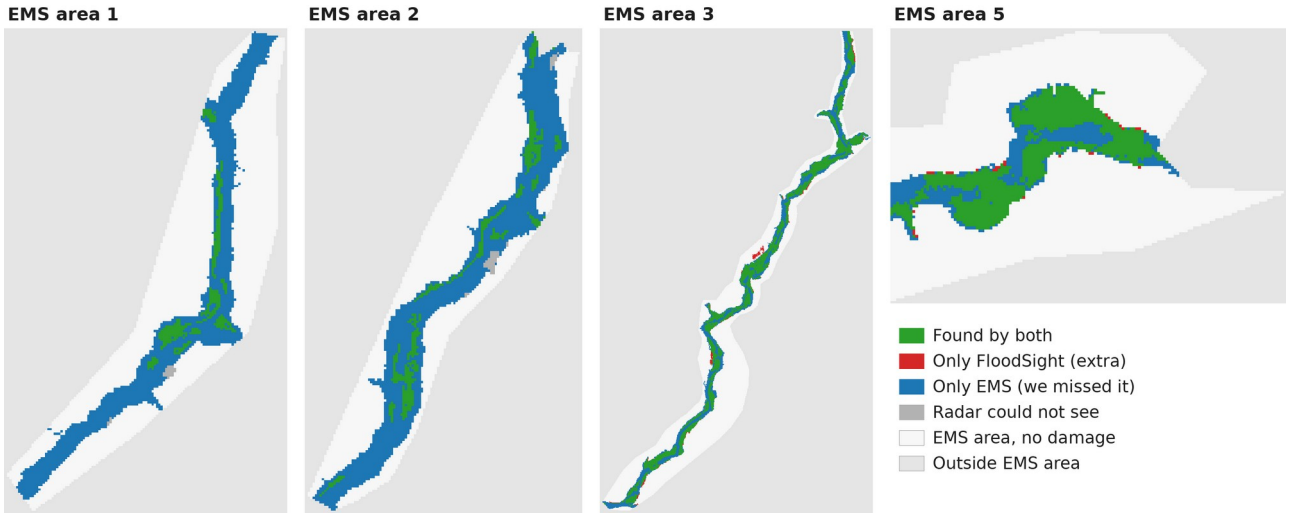


Figure 5. Trishuli case study (not blind): FloodSight's final map against the EMSR927 mass-movement maps, per EMS area. Green: both; red: only FloodSight; blue: only EMS (missed); grey: radar could not see. Precision is high (little red); the misses (blue) are mostly debris on an already rough riverbed in EMS areas 1-2.

Trishuli result	Value	What it means for responders
Settlements cut off (confirmed closures)	28	No road to a hospital even counting only confirmed damage: plan air or foot access
Likely cut off	5	Route survives only if single-source detections are wrong: check first
Unknown	4	Last route crosses roads hidden from the radar
Closed road length confirmed by EMS (EMS area 3)	98%	Closures that drive the cut-off list are reliable
Radar-unknown share of the valley	0.6%	Two tracks see almost everything; the rest is shown grey, not 'safe'

6.2 The AI component

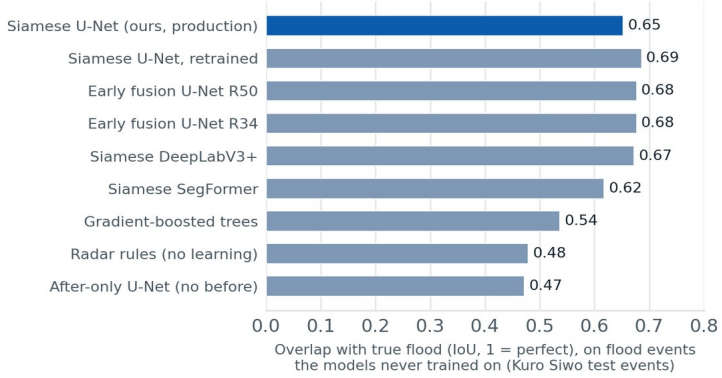


Figure 3. Eight alternatives trained with the same split and budget and scored on the same held-out Kuro Siwo tiles (flood IoU; each model's threshold chosen on the validation event). Ours (blue) was kept by the pre-registered validation rule.

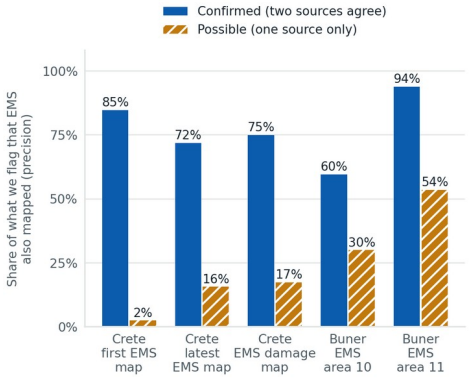


Figure 4. Precision of confirmed (two independent sources inside the system agree) versus possible (one source) detections, scored afterwards against Copernicus EMS maps.

Model comparison. Before training any alternative we fixed the set (Figure 3; early fusion is the Kuro Siwo benchmark's best family, the SegFormer follows ChangeFormer), the budget for every comparison model (same event split and test tiles, 15 epochs, batch 8, same loss, optimiser and augmentation, ImageNet start, thresholds chosen on validation only) and the rule: **a model replaces ours only if its validation IoU is 0.02 higher**. Our production model was trained earlier in its own run, so a same-budget retrain of the same architecture is included for a fair architecture comparison. The production model had the highest validation IoU (0.758; best challenger 0.726), so it stays. On the test events the two-date deep models lie within 0.07 of each other; the production model adds +0.17 IoU over the rules (0.653 vs 0.479) and +0.12 over gradient-boosted trees, and removing the before-image costs 0.18, which supports change detection. On the never-used event 205 the early-fusion models scored higher (0.61-0.63 vs 0.46); we report this as found and did not act on a test result.

AI alone on data it never trained on (table below; each row names its test type). On held-out Kuro Siwo events the production model reaches IoU 0.653 (F1 0.79), against a published benchmark best F1 of about 0.80 under a different setup. On Sen1Floods11, which has no before-image, the after-only variant scores 0.48 (test) and 0.68 (Bolivia), close to the dataset's per-chip Otsu labels (0.55 and 0.70) with one fixed setting. On the Himalayan flash flood it is 95-99% precise but finds under 1% of the area, because the damage was debris, not water: high precision with very low recall is not good detection, and the debris rules carry that case. On the blind Crete test it beats the rules alone against both EMS maps (0.55 vs 0.51 on the latest map, 0.84 vs 0.62 on the first).

AI alone (not the fused map)	Test type	IoU	Precision	Recall
Kuro Siwo events 427 + 1111006, same 600 tiles per event	Held-out events, same dataset	0.653	0.86	0.73
Kuro Siwo event 205, Sri Lanka	Blind held-out event	0.457	0.96	0.47
Sen1Floods11 test split, 90 chips (after-only variant, all water)	Second dataset	0.479	0.59	0.72
Sen1Floods11 Bolivia hold-out, 15 chips (after-only variant)	Second dataset	0.684	0.84	0.79
Buner EMS areas 10 / 11 vs EMS (debris-dominated)	Blind, Himalaya	0.006 / 0.006	0.95 / 0.99	0.006 / 0.006
Lasithi plateau vs EMS, latest / first map	Blind, 2026	0.549 / 0.842	0.74 / 0.89	0.68 / 0.94

Confidence is meaningful (Figure 4): confirmed detections are 60-94% precise against EMS, single-source detections 2-54%; Trishuli is the exception, with both above 0.94. This is why the dashboard and report lead with confirmed numbers and show the rest as possible.

7. Dashboard and situation report

The web app runs a new analysis from a drawn box and a date, shows every outside data request live (service, item, seconds, megabytes, processing units, waits for busy servers), and opens on a verdict with confirmed key numbers, a strip saying whether the area was checked against a reference map, and the settlements to reach first. Map views switch between the final map, the AI alone, the radar rules alone and a before/after slider; tabs cover cut-off settlements, the AI on this area, the flood path, the report, validation and method. A separate page documents how the model was tested. The **one-page situation report** is produced from the results by a fixed template, so no language model writes numbers; it is in English, plus Nepali for areas in Nepal, and lists blocked roads, bridges, cut-off settlements with travel times and air-access options. Responders also get CSV lists (settlements, hospitals) and GeoTIFF/GeoJSON layers. A public server lets judges run any area with an access code or their own free Copernicus keys, which are kept in memory for one run only.

7.1 Speed, cost and transparency

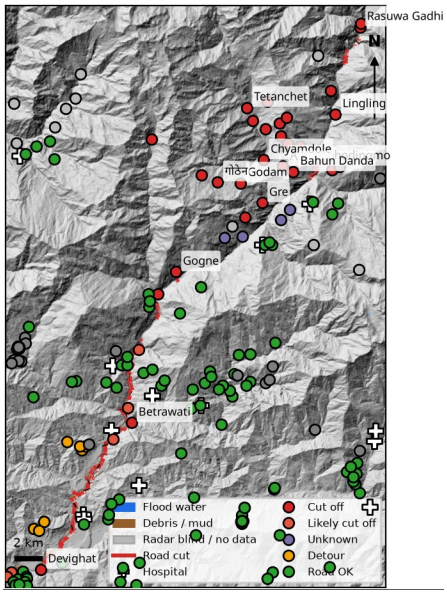


Figure 6. Situation-report map for the Trishuli run, as FloodSight produces it: road cuts on confirmed damage and every settlement coloured by access status; hospitals marked with crosses.

A 113 km² area (Lasithi) runs in 62 s with fresh downloads and 36 s when the images are cached, using about 24 processing units for four Sentinel-1 images and 26 for Sentinel-2 on the free Copernicus tier. Run time grows with area and with the number of mapped buildings and roads; damage and network steps run on all features at once, and OpenStreetMap layers download in parallel with automatic retry. Every step's time and every outside request is shown in the app, so a judge can see exactly where the time goes. 164 automated tests cover the data rules, detection, fusion, damage, access, reports and the public server.

8. Limitations and what we rejected

- **After the fact, not early warning:** Sentinel-1 revisits a track every 6-12 days; flash-flood water may have drained by the first pass.
- **Himalayan debris is often missed:** recall 48% (Trishuli), 34% and 10% (Buner); 30-42% of the missed Buner area showed no radar change at all. Strips of 20-60 m are 1-3 pixels at 20 m; 10 m gave no meaningful gain (Meža 0.014 to 0.020 for four times the cost).
- **The AI detects water, not debris,** and learned on flat terrain (0-2.5% steep ground vs 82% in Trishuli). Optical data is fused after the network because Kuro Siwo has no Sentinel-2.
- **Single-source detections over-flag** (Figure 4); they are therefore labelled possible. Cut-off results depend on OpenStreetMap completeness; 'cut off' means no mapped road route. Helicopter options are unvalidated candidates.
- **Our own GRD chain** uses the geolocation grid and a DEM shift instead of precise-orbit range-Doppler terrain correction.
- **Tested and not adopted** (pre-registered criteria not met): a median of four before-images as reference (Handwerger et al. 2022; seasonal change became false debris), four orbit tracks (no gain, twice the time), a learned normal-change test, smoothing the AI input. **Bugs found and fixed openly:** a track-merge overflow that let a blind track overwrite a seeing one (radar-unknown area in Trishuli 9.1% to 0.6%, all results re-run and both values logged), HAND undefined at the area edge, the model running on one track only, and inverted debris confidence: on Buner EMS area 10 'confirmed' debris was less precise than single-source debris (0.12 vs 0.46). Requiring a second source within 40 m raised it to 0.60 (EMS area 11: 0.50 to 0.94) with the map extent unchanged; on the Meža diagnosis it missed its pre-registered target, so debris confidence is not yet calibrated.

9. Reproducibility, data use and attributions

Code, configuration, tests and the full tuning log are in the GitHub repository; python run.py --bbox W,S,E,N --date YYYY-MM-DD reproduces any run, and an offline demo runs without an account. The pipeline uses only the allowed inputs: Copernicus EMS products (EMSR927, EMSR933, EMSR838, EMSR680) and Kuro Siwo or Sen1Floods11 labels serve only to score results. Contains modified Copernicus Sentinel data 2014-2026, processed on the Copernicus Data Space Ecosystem; Copernicus DEM GLO-30 (© DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018, provided under COPERNICUS by the European Union and ESA); © OpenStreetMap contributors (ODbL), via the ohsome API (HeiGIT); Copernicus Emergency Management Service (© European Union) for validation only; Natural Earth (public domain).

References. Bountos et al. 2024, Kuro Siwo, NeurIPS Datasets and Benchmarks. Bonafilia et al. 2020, Sen1Floods11, CVPR Workshops. Chini et al. 2017, IEEE TGRS (split-based flood thresholding). Kapos et al. 2000 (UNEP-WCMC mountain classes). Handwerger et al. 2022, NHESS 22, 753. Daudt et al. 2018 (FC-EF). Bandara and Patel 2022 (ChangeFormer). Xu 2006 (MNDWI). US Army FM 3-21.38 (landing-site slope).